For Examiner's Use

5 **(a)** Distinguish between electromotive force (e.m.f.) and potential difference (p.d.) using energy considerations.

.....

.....

.....

..... [2]

(b) A battery of e.m.f. 9.0 V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Figure 5.1.

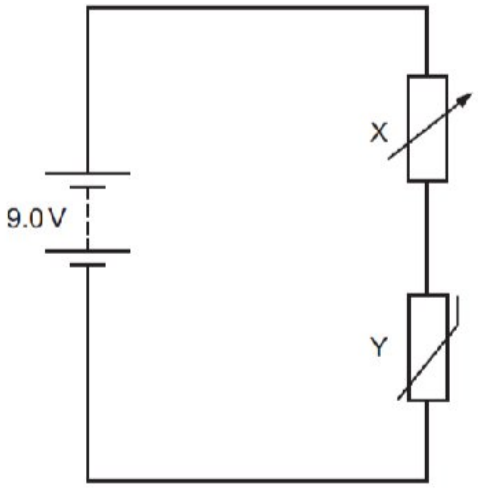


Figure 5.1

Figure 5.2 shows the variation with temperature of the resistance of the thermistor.

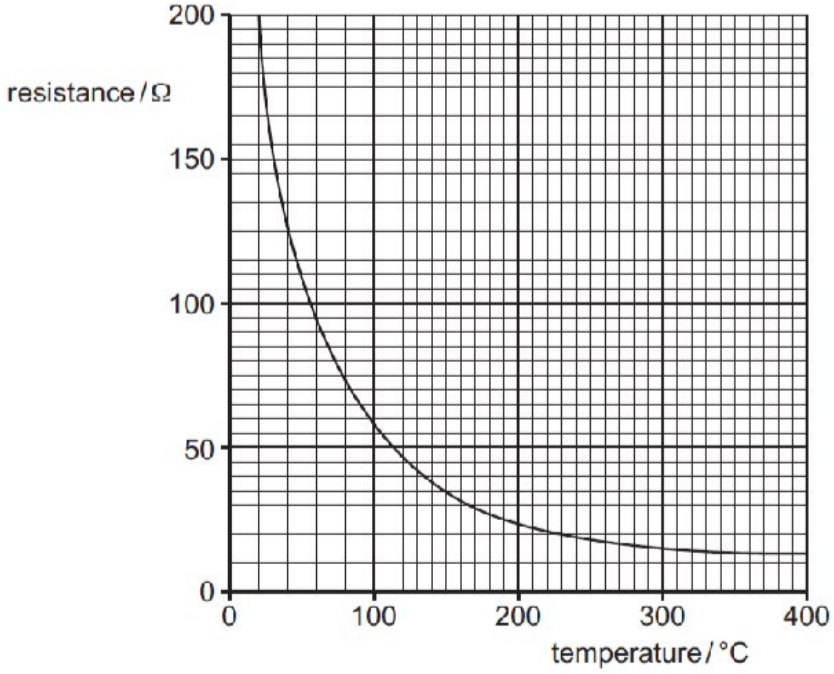


Figure 5.2

For Examiner's Use

(i) The current in the circuit is 0.011 A and the potential difference across Y is 4.0 V.

Calculate the resistance of X.

resistance = ... Ω [2]

(ii) The temperature of Y is changed to 190 °C. The resistance of X remains unchanged.

1. Determine the new potential difference across Y.

potential difference = ... V [3]

2. The resistance of X is increased.

State and explain how the value in **5(b)(ii)1.** will change.

.....

.....

.....

..... [2]

For Examiner's Use

(c) Figure 5.3 shows a network of capacitors connected to a 12 V supply. Z is a capacitor of capacitance Z.

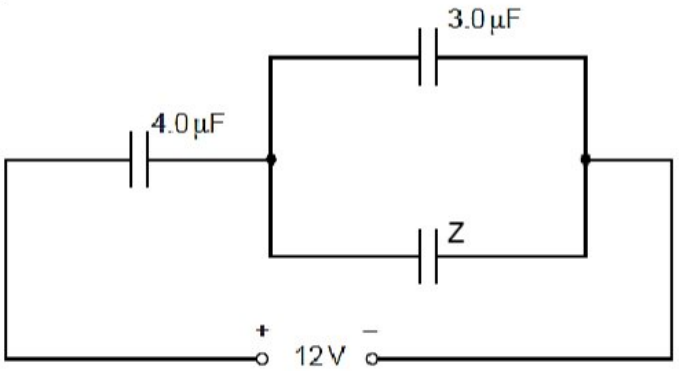


Figure 5.3

When the capacitors are fully charged, the potential difference across Z is 3.0 V.

(i) Calculate the charge stored in the 4.0 μF capacitor.

charge = ... C [2]

(ii) Determine the capacitance of Z.

capacitance = ... μF [3]

[Total: 14]